

# Supplementary Materials: Comprehensive Computational Data and Robustness Validations

This supplementary document provides the exhaustive numerical datasets underlying the phase transitions, algorithmic intervention validations (Hypotheses 1–3), and advanced robustness checks (Validations 1–4) discussed in the main text. The tables aggregate the integration outputs across macroscopic network scales ( $N \in [10^3, 10^7]$ ), ensuring the empirical transparency and theoretical reproducibility of all computational claims.

## S1. Autonomic Stability and Topological Hysteresis (Hypothesis 1)

To preempt critiques regarding the dependency of phase transitions on empirically calibrated initial conditions, the Simplicial TMFG model was strictly initialized utilizing a continuous uniform distribution,  $X_{i,0} \sim \mathcal{U}(0, 0.2)$ . Table S1 tracks the normalized proportion of ruptured 2-simplices ( $\Delta\beta_1/|F|$ ). During the unperturbed Burn-in phase ( $t = 49$ ), the defect ratio remains exactly at 0.000000 across all scales, proving autonomic self-organization without spontaneous failure. After pathogenic perturbation, the network locks into a permanently degraded hysteresis state.

| Table S1: Phase Evolution of Topological Defects ( $\Delta\beta_1/ F $ ) Prior to Intervention |                      |                       |                        |  |
|------------------------------------------------------------------------------------------------|----------------------|-----------------------|------------------------|--|
| Network Size ( $N$ )                                                                           | Phase 0 ( $t = 49$ ) | Phase 1a ( $t = 99$ ) | Phase 1b ( $t = 199$ ) |  |
| 1,000                                                                                          | 0.000000             | 0.062166              | 0.063503               |  |
| 10,000                                                                                         | 0.000000             | 0.025107              | 0.025907               |  |
| 100,000                                                                                        | 0.000000             | 0.007364              | 0.007624               |  |
| 1,000,000                                                                                      | 0.000000             | 0.002999              | 0.003095               |  |
| 10,000,000                                                                                     | 0.000000             | 0.000834              | 0.000859               |  |

## S2. Structural Failure of 1-Dimensional Interventions (Hypothesis 2)

Table S2 evaluates the structural response to classical 1-dimensional algorithmic interventions (e.g., edge recommendation systems). Injecting 1-simplices equivalent to 10% of the existing network topology deterministically exacerbates social fragmentation. Governed by the Euler-Poincaré formula, this operation uniformly induces an exact +0.200000 geometric penalty in the normalized void ratio, regardless of the network scale.

| Table S2: Defect Penalty Induced by 1-Dimensional Edge Injection |                      |                       |                            |  |
|------------------------------------------------------------------|----------------------|-----------------------|----------------------------|--|
| Network Size ( $N$ )                                             | Pre-1D ( $t = 199$ ) | Post-1D ( $t = 299$ ) | Defect Penalty Margin      |  |
| 1,000                                                            | 0.063503             | 0.263369              | +0.199866 $\approx$ +0.200 |  |
| 10,000                                                           | 0.025907             | 0.225894              | +0.199987 $\approx$ +0.200 |  |
| 100,000                                                          | 0.007624             | 0.207622              | +0.199998 $\approx$ +0.200 |  |
| 1,000,000                                                        | 0.003095             | 0.203095              | +0.200000 $\approx$ +0.200 |  |
| 10,000,000                                                       | 0.000859             | 0.200859              | +0.200000 $\approx$ +0.200 |  |

## S3. Efficacy of High-Dimensional Interventions (Hypothesis 3)

Table S3 quantifies the mathematical efficacy of high-dimensional interventions. The strategic injection of 2-simplices (triadic closures) at  $t = 300$  successfully bypasses the structural voids, shrinking the kernel of the Hodge Laplacian. Even starting from an uncalibrated uniform initial state, this operation achieves an instantaneous, scale-invariant reduction of approximately 95% of the accumulated  $\beta_1$  defects.

Table S3: Topological Recovery via 2-Simplex Injection

| Network Size ( $N$ ) | Pre-2D ( $t = 299$ ) | Post-2D ( $t = 300$ ) | Terminal State ( $t = 399$ ) |
|----------------------|----------------------|-----------------------|------------------------------|
| 1,000                | 0.081217             | 0.004345              | 0.004345                     |
| 10,000               | 0.031842             | 0.001600              | 0.001600                     |
| 100,000              | 0.008390             | 0.000420              | 0.000420                     |
| 1,000,000            | 0.002819             | 0.000141              | 0.000141                     |

#### S4. Optimal Control and Intervention Timing (Validation 1)

Table S4 quantifies the temporal trade-off in intervention execution. Delaying interventions geometrically escalates the requisite computational cost ( $M_{req}$ ). However, late interventions yield a mathematically superior terminal stability by completely exhausting pathogenic vulnerabilities prior to execution.

Table S4: Peak Cost Escalation and Terminal Efficacy in Optimal Control

| Size ( $N$ ) | Peak Required Cost ( $M_{req}$ ) |                    | Terminal Defect ( $t = 399$ ) |            |
|--------------|----------------------------------|--------------------|-------------------------------|------------|
|              | Early ( $t = 129$ )              | Late ( $t = 179$ ) | Early Final                   | Late Final |
| 1,000        | 0.061163                         | 0.074866           | 0.002340                      | 0.000000   |
| 100,000      | 0.005890                         | 0.007700           | 0.000187                      | 0.000037   |
| 1,000,000    | 0.001846                         | 0.002377           | 0.000058                      | 0.000010   |

#### S5. Topological Side Effects: Cult Formation (Validation 2)

Table S5 evaluates the structural side effects induced by algorithmic interventions. Forcing the absolute eradication of social voids ( $\beta_1 \rightarrow 0$ ) necessitates the spatial saturation of 2-simplices, which combinatorially synthesizes closed 3-simplices ( $\beta_2$ , mapping to isolated cultic echo-chambers).

Table S5: Quantitative Trade-off: Mitigating Voids ( $\beta_1$ ) vs. Generating Cults ( $\beta_2$ )

| Size ( $N$ ) | Pre-Intervention ( $t = 299$ ) |                   | Post-Intervention ( $t = 300$ ) |                   |
|--------------|--------------------------------|-------------------|---------------------------------|-------------------|
|              | $\beta_1$ (Voids)              | $\beta_2$ (Cults) | $\beta_1$ (Voids)               | $\beta_2$ (Cults) |
| 1,000        | 0.077874                       | 0.000000          | 0.000000                        | 0.032754          |
| 1,000,000    | 0.002451                       | 0.000000          | 0.000000                        | 0.000030          |

#### S6. Time-Lagged Rupture in Adaptive Networks (Validation 3)

Table S6 confirms the temporal divergence between consensus collapse and connection severing. During the initial shock phase ( $t = 105$ ), 2-simplex face ruptures ( $\beta_1$ ) exhibit a massive spike while 1-simplex edge severing ( $\beta_0$ ) remains mathematically negligible, confirming a deterministic window of opportunity for interventions.

Table S6: Temporal Progression of Face Rupture vs. Edge Severing

| Size ( $N$ ) | Initial Shock ( $t = 105$ ) |                       | Mid Pathogenesis ( $t = 150$ ) |                    |
|--------------|-----------------------------|-----------------------|--------------------------------|--------------------|
|              | $\beta_1$ Rupture           | $\beta_0$ Severing    | $\beta_1$ Rupture              | $\beta_0$ Severing |
| 1,000        | 0.055816                    | 0.000167              | 0.060495                       | 0.043921           |
| 1,000,000    | 0.001659                    | $1.66 \times 10^{-7}$ | 0.001763                       | 0.001298           |

## S7. Universal Robustness under Uncalibrated Nodal Heterogeneity (Validation 4)

A standard critique in computational social science is that heterogeneous agent-based models demand rigorous empirical calibration to prevent arbitrary phase transitions. To preemptively counter this, Validation 4 assigns nodal frustration thresholds from a statistical distribution ( $\theta_i \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$ ) without applying any ex-post adjustments to the network topology.

Table S7 demonstrates the mathematical robustness of the model under these strict conditions. Due to the hypersensitivity of outlier nodes in the threshold distribution, the system autonomously digests the initial noise to form a stable, non-zero baseline ( $t = 99$ ). Despite this structurally complex heterogeneous state, the high-dimensional intervention successfully compresses the elevated peak damage ( $t = 299 \rightarrow 399$ ). This definitively proves that the recovery mechanics (kernel shrinkage of the Hodge Laplacian) are universal geometric properties of scale-free simplicial complexes, entirely independent of parameter tuning or behavioral homogenization.

Table S7: Recovery Robustness under Uncalibrated Heterogeneous Constraints

| Network Size ( $N$ ) | Stationary ( $t = 99$ ) | Peak Damage ( $t = 299$ ) | Recovered ( $t = 399$ ) |
|----------------------|-------------------------|---------------------------|-------------------------|
| 1,000                | 0.012701                | 0.131350                  | 0.007687                |
| 10,000               | 0.012503                | 0.067385                  | 0.005501                |
| 100,000              | 0.016137                | 0.041621                  | 0.004333                |
| 1,000,000            | 0.016129                | 0.033089                  | 0.004151                |

## Conclusion and Synthesis of Computational Validations

This study established the Simplicial Topological Mean Field Game (TMFG) model to deterministically evaluate the mechanics of consensus collapse and algorithmic recovery in scale-free social networks. The numerical integration across macroscopic scales up to  $N = 10^7$  nodes mathematically validates all proposed hypotheses and demonstrates the systemic robustness of the high-dimensional intervention framework.

First, under strictly uncalibrated uniform initial conditions ( $X_{i,0} \sim \mathcal{U}(0, 0.2)$ ), the system autonomously self-organized without spontaneous failure ( $\Delta\beta_1/|F| = 0.000000$  at  $t = 99$ ). The subsequent introduction of pathogenic conformity pressure induced a structural phase transition, permanently degrading the network geometry into a hysteresis state (0.000859 for  $N = 10^7$ ). This verifies **Hypothesis 1**: localized information gradients rupture 2-simplices to create irreversible topological porosity, establishing a deterministically stable baseline independent of empirical parameter tuning.

Second, the computational results explicitly confirmed **Hypothesis 2** regarding the geometric failure of 1-dimensional interventions. Injecting 1-simplices equivalent to 10% of the existing network topology deterministically induced a uniform +0.200000 penalty in the normalized void ratio across all evaluated scales. Governed by the Euler-Poincaré formula ( $\Delta\beta_1 =$

$\Delta\beta_0 + \Delta|E| - \Delta|F|$ ), this proves that standard edge recommendation algorithms act as structural catalysts that accelerate, rather than mitigate, macroscopic fragmentation.

Third, **Hypothesis 3** was fully corroborated through the strategic application of 2-simplex injections. The algorithmic construction of triadic closures successfully shrank the kernel of the Hodge Laplacian. This high-dimensional bypass operation achieved an instantaneous, scale-invariant reduction of accumulated  $\beta_1$  defects by approximately 95% (e.g., compressing the defect ratio from 0.002819 down to 0.000141 for  $N = 10^6$ ).

Furthermore, the advanced validations confirmed the operational viability of the model under complex systemic constraints:

- **Temporal Window of Opportunity (Validation 3):** Adaptive rewiring dynamics confirmed that 2-simplex face ruptures strictly precede 1-simplex edge severing. At  $t = 105$  for  $N = 10^6$ ,  $\beta_1$  damage spiked to 0.001659 while physical  $\beta_0$  disconnection remained mathematically negligible at  $1.66 \times 10^{-7}$ . This proves the existence of a deterministic temporal lag, allowing for interventions prior to absolute network segregation.
- **Optimal Control and Side Effects (Validations 1 & 2):** The framework identified a strict therapeutic trade-off. Delayed interventions exhaust pathogenic vulnerabilities to yield mathematically superior terminal stability (defect ratio 0.000010 vs 0.000058). However, forcing the absolute eradication of  $\beta_1$  voids combinatorially necessitates spatial saturation. This saturation synthesizes closed 3-simplices, inflicting a permanent  $\beta_2$  cultivation penalty (0.000030 for  $N = 10^6$ ).
- **Robustness to Heterogeneity (Validation 4):** The recovery mechanics remained fundamentally robust under severe, uncalibrated nodal heterogeneity ( $\theta_i \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$ ). Despite establishing an elevated non-zero stationary baseline (0.016129 for  $N = 10^6$ ), the intervention successfully compressed the peak pathogenesis (0.033089) down to a stable recovered state (0.004151). This confirms that the model’s efficacy is a universal geometric property of simplicial complexes.

In summation, this research proves that information pollution in digital platforms is fundamentally a higher-order topological phenomenon. Combating this pathology requires transitioning algorithmic governance from classical graph-theoretic edge recommendations to the optimal control of simplicial topologies. The defined Simplicial TMFG framework provides a rigorous, scale-invariant mathematical foundation for executing these high-dimensional bypass surgeries while strictly bounding their structural side effects.